

Name:	
Class:	



YEAR 12
ASSESSMENT TEST 2
TERM 1, 2013

MATHEMATICS

Time Allowed – 90 Minutes
(Plus 5 minutes Reading Time)

General Instructions:

- All questions may be attempted
- All questions are of equal value
- Standard Integral Tables will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

The answers to all questions are to be returned in separate bundles clearly labelled Question 1, Question 2, etc. Each question must show your Candidate Number.

QUESTION 1 15 Marks

Marks

(a) Find the following indefinite integrals:

(i) $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right) dx$ 2

(ii) $\int \frac{x^2 + 2x - 1}{x^2} dx$ 2

(iii) $\int x e^{3x^2} dx$ 1

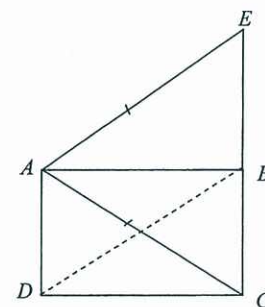
(iv) $\int \cos \pi x dx$ 1

(b) A rectangle measures 64 cm by 2 cm. The rectangle is changed in size, so that the length is decreased by $33\frac{1}{3}\%$ and the width is increased by $33\frac{1}{3}\%$.

(i) Find the ratio of the new area of the rectangle to the original area. 2

(ii) How many times must this process be performed so that the new shape is similar in shape to the original one? 2

(c) $ABCD$ is a rectangle. CB is produced to E , so that $AE = AC$.



Not to Scale

(i) Prove that $\triangle ABE \cong \triangle ABC$, giving reasons. 3

(ii) Prove that $AEBD$ is a parallelogram, giving reasons. 2

QUESTION 2 15 Marks START A NEW PAGE

Marks

- (a) Sketch, on a number plane, the graph of $y = \frac{2-x}{x+1}$.
Label all intercepts and asymptotes. 3
- (b) Three **different** positive numbers, 2, x , y are the first, second and twelfth terms of an arithmetic sequence. They are also consecutive terms of a geometric sequence.
- (i) Show that $y = 11x - 20$. 2
- (ii) Find the values of x and y . 3
- (c) The line $y = \frac{1}{2}$ intersects the curve $y = \sin x$, in the domain $0 \leq x \leq 2\pi$,
at two points $A\left(\frac{\pi}{6}, \frac{1}{2}\right)$ and $B\left(\frac{5\pi}{6}, \frac{1}{2}\right)$.
- (i) Calculate the area bounded by the curve $y = \sin x$ and the line $y = \frac{1}{2}$,
between points A and B . 3
- (ii) This area is rotated about the x -axis. Use the trapezoidal rule with 5 function values
to find an approximation to the volume of the solid of revolution formed.
Give your answer to 2 decimal places. 4

QUESTION 3 15 Marks START A NEW PAGE

- (a) The first four terms of a sequence are:

$$1, \frac{1+2}{1+3}, \frac{1+2+3}{1+3+5}, \frac{1+2+3+4}{1+3+5+7}.$$

- (i) Find a formula for T_n , the n th term of the sequence. 2
- (ii) Show that T_n is always greater than $\frac{1}{2}$. 1
- (b) The area bounded by the x and y axes, the line $y = 1$ and the graph $y = \ln x$ is rotated
about the y axis. Calculate the volume of the solid of revolution formed. 3
- (c) Use Simpson's rule with 5 function values to estimate the area bounded by the
the graph $y = -\sqrt{25-x^2}$, the x and y axes and the line $x = 4$.
Give your answer to 2 decimal places. 4

Question 3 continued over page

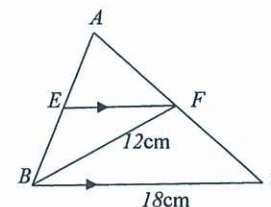
Question 3 continued

Marks

- (d) Anna opens a bank account to save for the deposit to buy a house. At the beginning of each month she makes a deposit into the account. Her first deposit is \$100. Each month she increases her deposit by 1%. The interest rate on the account is 6% p.a., paid monthly.
- (i) Show that the amount in her account, A_n at the end of n months,
is given by the formula: 3
$$A_n = 20100(1.01^n - 1.005^n).$$
- (ii) Calculate the total amount of interest, to the nearest dollar, that Anna received
on her investment at the end of 10 years. 2

QUESTION 4 15 Marks START A NEW PAGE

- (a) Show on a number plane the region representing the set: 3
 $\{(x, y) : y \leq -3x + 6\} \cap \{(x, y) : (x-1)^2 + (y+2)^2 < 16\}$
- (b) The first term of an infinite geometric series is 6. Each term is double the sum of all the terms that follow.
Find the values of the common ratio and the limiting sum of the series. 3
- (c) In $\triangle ABC$, $\angle ABC = 2\angle ACB$. BF bisects $\angle ABC$ and EF is parallel to BC .
 $BC = 18$ cm and $BF = 12$ cm.



Not to Scale

- (i) Prove that $\triangle EBF$ and $\triangle FBC$ are similar isosceles triangles, giving reasons. 3
- (ii) Show that the length of EB is 8 cm, giving reasons. 2
- (iii) Calculate the length of AF , giving reasons. 4

END OF EXAMINATION

MATHEMATICS: Question 1

Suggested Solutions	Marks	Marker's Comments
a) i) $\int (\sqrt{x} - \frac{1}{\sqrt{x}}) dx = \int (x^{\frac{1}{2}} - x^{-\frac{1}{2}}) dx$ $= \frac{2}{3} x^{\frac{3}{2}} - 2x^{\frac{1}{2}} + c$ $= \frac{2}{3} x\sqrt{x} - 2\sqrt{x} + c$	1	Convert to index form
	1	Integrate
ii) $\int \frac{x^2 + 2x - 1}{x^2} dx = \int (1 + \frac{2}{x} - \frac{1}{x^2}) dx$ $= x + 2\ln x + \frac{1}{x} + c$	1	Divide through
	1	Integrate
iii) $\int x e^{3x^2} dx = \frac{1}{6} e^{3x^2} + c$	1	Integrate
iv) $\int \cos \pi x dx = \int -1 dx$ $= -x + c$	$\frac{1}{2}$	Evaluate $\cos \pi$
	$\frac{1}{2}$	Integrate
b). Let L_n, W_n & A_n be the length, width & area (respectively) after the nth adjustment of dimensions. i) $\therefore L_0 = 64, W_0 = 2, A_0 = 128$ $L_1 = 64 \times \frac{2}{3}, W_1 = 2 \times \frac{4}{3}$ $= \frac{128}{3}, = \frac{8}{3}$ $\therefore A_1 = \frac{1024}{9}$ $\therefore \text{Ratio of areas} = \frac{(\frac{1024}{9})}{128}$ $= \frac{8}{9} \text{ (or } 8:9)$	$\frac{1}{2}$	Original area
	$\frac{1}{2}$	New area
ii) For similar shape, proportion of width to length must reverse ie $\frac{W_n}{L_n} = \frac{L_0}{W_0} \quad \left[\frac{(\frac{4}{3})}{(\frac{2}{3})} \right]^n = 1024$ $\frac{2(\frac{4}{3})^n}{64(\frac{2}{3})^n} = \frac{64}{2} \quad 2^n = 1024$ $\therefore n = 10$	1	Produce correct equation to be solved
	1	Number of times

MATHEMATICS: Question 1

Suggested Solutions	Marks	Marker's Comments
c.) i.) In $\triangle ABE$ & $\triangle ABC$: AB is common $AE = AC$ (given) $\angle ABC = 90^\circ$ (angle of rect- angle $ABCD$ is 90°) $\angle ABE + \angle ABC = 180^\circ$ (angle sum of straight angle is 180°) $\therefore \angle ABE = 90^\circ$ $\therefore \triangle ABE \equiv \triangle ABC$ (RHS)	1 1 $\frac{1}{2}$ $\frac{1}{2}$	Corresponding sides Corresponding angles Setting out to show equivalence of angles Correct reason for congruence
ii.) $AD \parallel BC$ (opposite sides of rectangle $ABCD$ are parallel) $\therefore AD \parallel BE$ (BE is extended from BC) $BC = BE$ (corresponding sides of congruent $\triangle ABE$ & $\triangle ABC$ equal) But $BC = AD$ (opposite sides of rectangle $ABCD$ are equal) $\therefore AD = BE$ $\therefore ADBE$ is a parallelogram (one pair of sides both equal & parallel)	$1\frac{1}{2}$ $\frac{1}{2}$	Sound reasoning that corresponds to relevant geometric features & properties Valid sufficiency condition chosen

1/2

MATHEMATICS: Question 2

Suggested Solutions

Marks

Marker's Comments

(a) $y = \frac{2-x}{x+1}$

vertical asymptote at $x = -1$
horizontal asymptote at $y = -1$

(2, 0) (0, 2)

$x = -1$

$y = -1$

$x = -1$

$y = -1$

$x = -1$

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$x = -1$

$y = -1$

$x = -1$

$y = -1$

problems

* inconsistent scale on axes.

* didn't lock a point in on 2nd branch.

* forgot horizontal asymptote.

-1/2 off for each error.

b(i) $T_1 = 2$ $T_2 = x$ $T_3 = y$

$d = T_2 - T_1 = x - 2$ $T_1 = 2 = a$

$T_{12} = a + 11d$
 $= 2 + 11(x - 2)$
 $\therefore y = 11x - 20$

(ii) A.P. $T_1 = 2$ $T_2 = x$ $T_3 = y$

$\therefore r = \frac{x}{2} = \frac{y}{x}$ $\therefore x^2 = 2y$

sub into $y = 11x - 20$
 $x^2 = 11x - 20$
 $x^2 - 11x + 20 = 0$
 $(x - 5)(x - 4) = 0$

$d = x - 2 = \frac{1}{2}mk$

1/2

1/2

1/2

1/2

1/2

1/2

1/2

1/2

a lot forgot to make the 40 positive

2/2

MATHEMATICS: Question..2... (cont)

Suggested Solutions

Marks

Marker's Comments

$$\therefore x=20 \text{ or } x=2$$

$$\text{when } x=20, y=200$$

$$\text{when } x=2, y=2 \text{ but } x \neq y$$

$$\therefore x=20, y=200 \text{ only}$$

1/2

1/2 off for each error

$$(5) (i) A = \int_{\pi/6}^{5\pi/6} (\sin x - \frac{1}{2}) dx$$

$$= \left[-\cos x - \frac{1}{2}x \right]_{\pi/6}^{5\pi/6}$$

$$= \left(-\cos \frac{5\pi}{6} - \frac{5\pi}{12} \right) - \left(-\cos \frac{\pi}{6} - \frac{\pi}{12} \right)$$

$$= \frac{\sqrt{3}}{2} - \frac{5\pi}{12} + \frac{\sqrt{3}}{2} + \frac{\pi}{12}$$

$$= \left(\sqrt{3} - \frac{\pi}{3} \right) \text{ units}^2$$

* maximum 2 mks if $\sqrt{3}$ or 0.684853 is the answer (no working)

$$(ii) f(x) = \sin x - \frac{1}{2}$$

$$V = \pi [f(x)]^2$$

$$= \pi \sin^2 x - \pi \left(\frac{1}{2} \right)^2$$

5 ln values = 4 applications

x	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$
f(x)	$\frac{1}{2} - \frac{1}{2}$	$\frac{\sqrt{3}}{2} - \frac{1}{2}$	$\frac{1}{2} - \frac{1}{2}$	$-\frac{\sqrt{3}}{2} - \frac{1}{2}$	$-\frac{1}{2} - \frac{1}{2}$
	0	$= \frac{2\sqrt{3}}{4}$	$\frac{3\sqrt{3}}{4}$	$-\frac{2\sqrt{3}}{4}$	0

* majority of students didn't write out f(x), so I had no idea what they were substituting into!!!

* A lot of students forgot to multiply by π .

$$V = \frac{\pi}{2} \left[f\left(\frac{\pi}{6}\right) + 2f\left(\frac{2\pi}{6}\right) + 2f\left(\frac{3\pi}{6}\right) + 2f\left(\frac{4\pi}{6}\right) + f\left(\frac{5\pi}{6}\right) \right]$$

$$= \frac{\pi}{2} \left[0 + 2\left(\frac{2\sqrt{3}}{4}\right) + 2\left(\frac{3\sqrt{3}}{4}\right) + 2\left(-\frac{2\sqrt{3}}{4}\right) + 0 \right]$$

$$= \frac{\pi}{2} \left(\frac{4\sqrt{3} + 6\sqrt{3} + 4\sqrt{3}}{4} \right)$$

$$= \frac{\pi}{2} \cdot \frac{7\sqrt{3}}{2}$$

$$= \frac{7\pi^2}{24}$$

$$= 2.878634617$$

$$\therefore V = 2.88 \text{ units}^3$$

* Some students found the area, lost 1 mk.

* 1/2 mk off if you squared it together instead of individually

* -1/2 if not to 2dp.

* 1/2 off for each error

Suggested Solutions

Marks

Marker's Comments

$$3(a)(i) \quad T_n = \frac{1+2+3+4+\dots+n}{1+3+5+7+\dots+2n-1}$$

$$\text{Let } S_n = 1+2+3+4+\dots+n \quad \text{AP}$$

$$a=1, d=1, S_n = \frac{n}{2}(1+n)$$

$$\text{Let } R_n = 1+3+5+7+\dots+2n-1 \quad \text{AP}$$

$$a=1, d=2, R_n = \frac{n}{2}(1+2n-1) \\ = n^2$$

$$\therefore T_n = \frac{\frac{n}{2}(1+n)}{n^2}$$

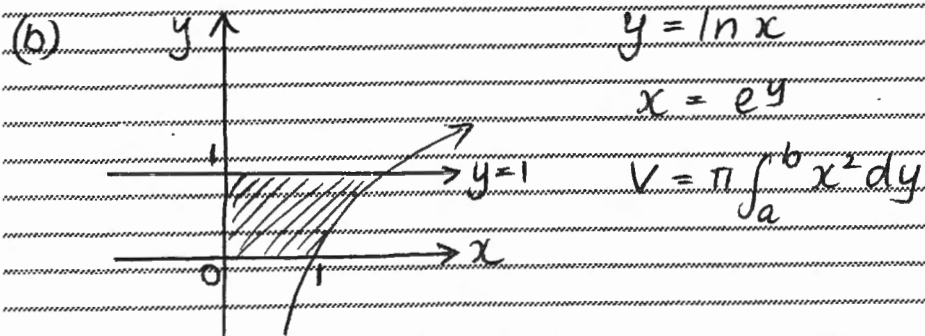
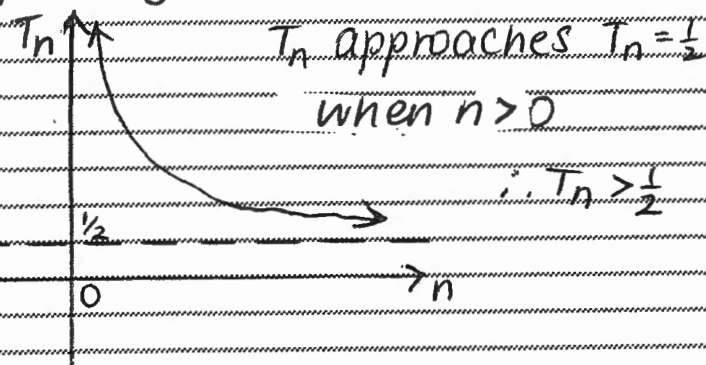
$$= \frac{(1+n)}{2n}$$

$$(ii) \quad T_n = \frac{1}{2n} + \frac{1}{2}$$

$$\text{Since } n > 0, \frac{1}{2n} > 0$$

$$\therefore T_n > \frac{1}{2}$$

OR, graphically:



1 numerator sum
1 denominator sum

1/2 Splitting T_n

1/2 reason

1 drawing graph
with
explanation

$$V = \pi \int_0^1 e^{2y} dy$$

$$= \pi \left[\frac{1}{2} e^{2y} \right]_0^1$$

$$= \pi \left[\frac{1}{2} e^2 - \frac{1}{2} e^0 \right]$$

$$= \pi \left[\frac{e^2}{2} - \frac{1}{2} \right]$$

$$= \frac{\pi}{2} (e^2 - 1) \text{ units}^3$$

1

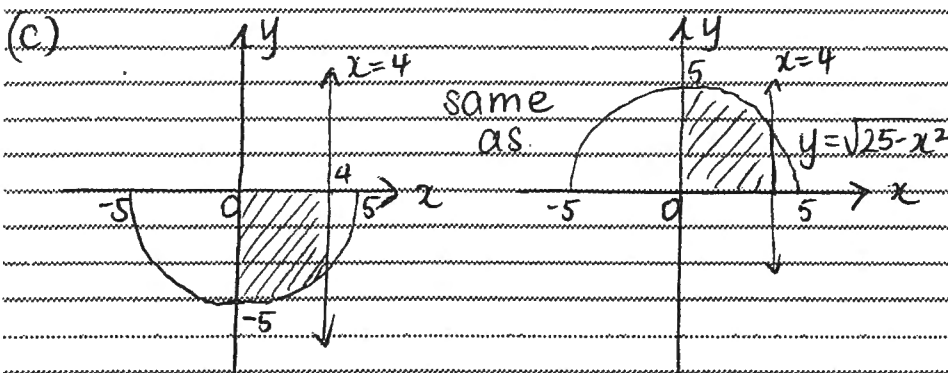
Integral

1

integration and substitution

1

answer with working.



x	0	1	2	3	4
y	5	$\sqrt{24}$	$\sqrt{21}$	4	3

1 1/2

Table values

$$A = \frac{1}{3} [5 + 4(\sqrt{24} + 4) + 2(\sqrt{21}) + 3]$$

$$\approx 17.58702311$$

$$\approx 17.59 \text{ units}^2 \text{ (to 2 dec. pl.)}$$

2

substitution into correct formula

1/2

answer

OR

$$A_1 = \frac{2-0}{6} [5 + 4\sqrt{24} + \sqrt{21}]$$

$$A_2 = \frac{4-2}{6} [\sqrt{21} + 4 \times 4 + 3]$$

$$\text{Total area} \approx 17.58702311$$

$$\approx 17.59 \text{ u}^2$$

1

substitution into correct

1

formula

1/2

answer

$$(d) (i) A_1 = 100 (1.005)$$

$$A_2 = 100 (1.005)^2 + 100 (1.01) (1.005)$$

$$A_3 = 100 (1.005)^3 + 100 (1.01) (1.005)^2 + 100 (1.01)^2 (1.005)$$

$$A_n = 100 (1.005)^n + \dots + 100 (1.01)^{n-1} (1.005)$$

$$A_n = 100 (1.005)^n \left[\frac{(1.01)^n - 1}{1.005 - 1} \right]$$

$$= 100 \left[\frac{1.01^n - 1.005^n}{1.005 - 1} \right]$$

$$= \frac{100 (1.005)}{0.005} [1.01^n - 1.005^n]$$

$$= 20100 [1.01^n - 1.005^n]$$

$$(ii) A_{120} = 20100 (1.01^{120} - 1.005^{120})$$

$$= \$29767.90$$

Total invested : $a=100, r=1.01, n=120$

$$S_{120} = \frac{100 (1.01^{120} - 1)}{1.01 - 1}$$

$$= \$23003.87$$

$$\text{Interest} = A_{120} - S_{120}$$

$$= \$6764 \text{ to nearest dollar}$$

1

series

1

sum

1

simplifying

$\frac{1}{2}$

A_{120}

1

S_{120}

$\frac{1}{2}$

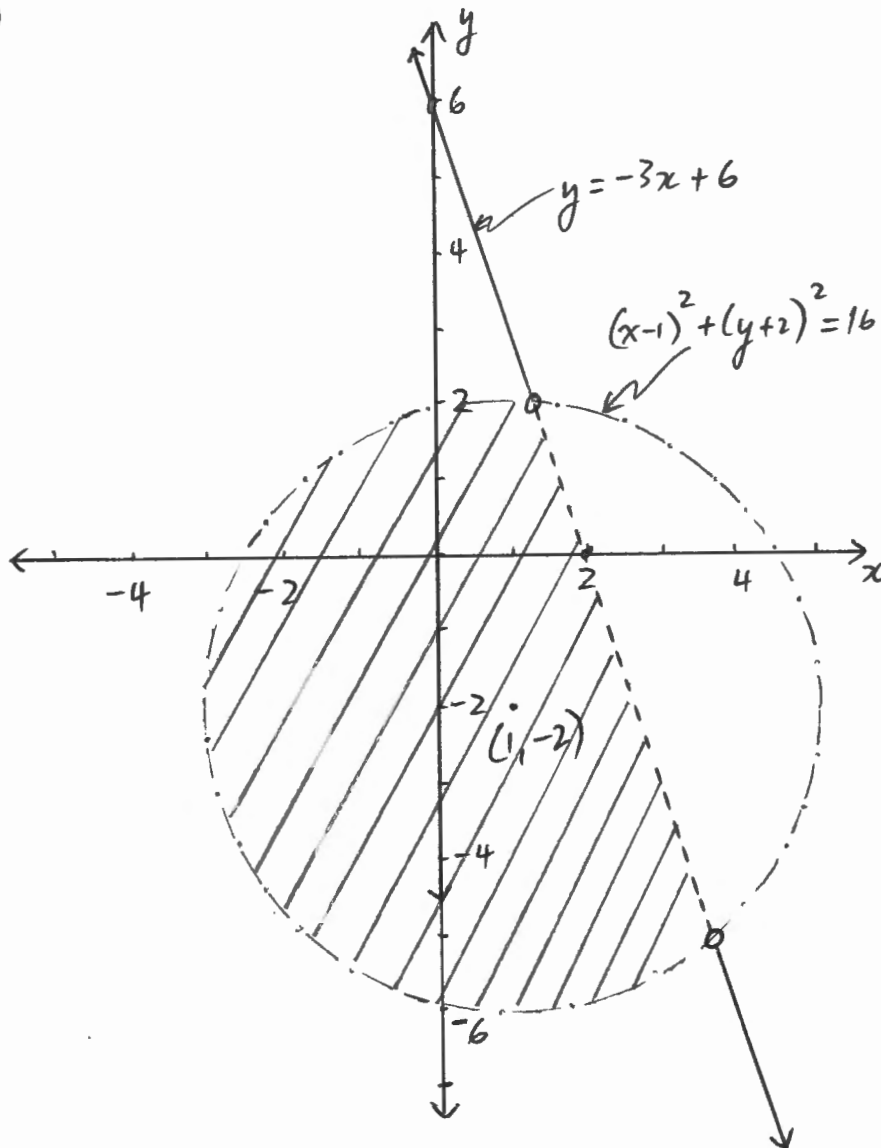
Interest

Suggested Solutions

Marks

Marker's Comments

a)

 $\frac{1}{2}$

for st. line

 $\frac{1}{2}$

for doty line

 $\frac{1}{2}$ for circle centre
(1, -2) and $r=4$ $\frac{1}{2}$

for doty line

 $\frac{1}{2}$ for open ring
at points of
intersection $\frac{1}{2}$

shaded region

 $-\frac{1}{2}$ for not
labelling graphs $-\frac{1}{2}$ for not using
compass $-\frac{1}{2}$ for incorred/
poor scale $-\frac{1}{2}$ for no centre

Suggested Solutions	Marks	Marker's Comments
b) $T_n = 2(S_\infty - T_{n+1})$ $\therefore 6 = 2(S_\infty - 6)$ $\Rightarrow S_\infty = 9$ i.e. $\frac{a}{1-r} = 9$ $\therefore 1-r = \frac{6}{9}$ $\therefore r = \frac{1}{3}$ Hence common ratio is $\frac{1}{3}$ and the limiting sum is 9.	1 1 1	$S_\infty = \frac{a}{1-r}$ (formula) $S_\infty = 9$ $r = \frac{1}{3}$
c) (i) In Δ's EBF and FBC • Let $\angle EBF = x^\circ$ - Now $\angle EBF = \angle FBC = x \dots$ (FB bisects $\angle ABC$) • $\angle EFB = \angle FBC \dots$ (alternate angles on parallel lines EF and BC, are equal) But $\angle FCB = \frac{1}{2} \angle ABC \dots$ (given) $= x$ $\therefore \angle FCB = \angle EBF \dots$ (from •) $\therefore \Delta EBF \parallel \Delta FBC \dots$ (equiangular) But $\angle FBC = \angle FCB \dots$ (proved above) Hence ΔFBC is isosceles $\dots$ (base angles are equal)	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	for $\angle EBF = \angle FBC$ for isosceles ΔFBC

Suggested Solutions	Marks	Marker's Comments
But $\triangle FBC$ is similar to $\triangle EBF$ and since $\triangle FBC$ is isosceles, $\triangle EBF$ must be isosceles.	$\frac{1}{2}$	isosceles $\triangle EBF$
(ii) From $\triangle EBF \parallel \triangle FBC$, $\frac{EB}{FB} = \frac{BF}{BC} \dots \text{(corresponding sides in similar } \triangle\text{'s are in proportion)}$ i.e. $\frac{EB}{12} = \frac{12}{18}$ $\therefore EB = 12^2/18$ $= 8 \text{ cm.}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	ratio correct for reason ^ for sub.
(iii) $EB = EF = 8 \text{ cm} \dots (\triangle EBF \text{ isosceles})$ $FC = 12 \text{ cm} \dots (\triangle FBC \text{ isosceles})$ In $\triangle AEF$ & $\triangle ABC$, $\angle EAF$ is common $\angle ABC = \angle AEF \dots$ (corresponding angles on parallel lines, EF and BC are equal) $\therefore \triangle AEF \parallel \triangle ABC \dots$ (equiangular) Hence $\frac{AF}{AC} = \frac{FE}{BC} \dots$ (corresponding sides on parallel sides EF and BC are equal) But $AC = AF + FC$ $= AF + 12$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 1	final correct answer for $EF = 8 \text{ cm}$ $FC = 12 \text{ cm}$ with reason Similar $\triangle$'s correct ratio

Suggested Solutions

Marks

Marker's Comments

$$\text{Hence } \frac{AF}{AF+12} = \frac{8}{18}$$

↳ Solving for AF gives 9.6 cm

$$\text{i.e. } AF = \underline{9.6 \text{ cm}} \rightarrow$$

1

for correct
answer